

Exam 1 Review: Scalar Flows and Bifurcations

Tuesday, October 6, 2026

Idea: Scalar-flow analysis

For a parameter-dependent scalar equation

$$\dot{x} = f(x; r),$$

use the chain

$$\boxed{\text{state space}} \longrightarrow \boxed{f(x; r) = 0} \longrightarrow \boxed{\text{sign } f} \longrightarrow \boxed{\text{stability and limits}} \longrightarrow \boxed{\text{parameter dependence}}.$$

Recall: Hyperbolic equilibrium test

Let x^* be an equilibrium of $\dot{x} = f(x; r)$, where f is continuously differentiable with respect to x . If $f_x(x^*; r) \neq 0$, then x^* is **hyperbolic**, and

$$f_x(x^*; r) < 0 \Rightarrow x^* \text{ is asymptotically stable,} \quad f_x(x^*; r) > 0 \Rightarrow x^* \text{ is unstable.}$$

Observation: Nonhyperbolic scalar equilibria

For $\dot{x} = f(x; r)$, if $f_x(x^*; r) = 0$, the derivative test is inconclusive. When x^* is isolated, use the signs of $f(x; r)$ on the intervals immediately to its left and right to determine stability.

Question: What is the state space?

Classify each equation as autonomous or nonautonomous and linear or nonlinear. Give an autonomous state space.

$$\dot{x} = x(1 - x), \quad \ddot{q} + q^3 = 0, \quad \dot{y} = t - y.$$

Solution.

equation	autonomous	linearity	autonomous state
$\dot{x} = x(1 - x)$	yes	nonlinear	$x \in \mathbb{R}$
$\ddot{q} + q^3 = 0$	yes	nonlinear	$(q, v) \in \mathbb{R}^2$
$\dot{y} = t - y$	no	linear nonhomogeneous	$(y, s) \in \mathbb{R}^2$

For the second and third equations,

$$\dot{q} = v, \quad \dot{v} = -q^3,$$

Question: What is the state space? (continued)

and

$$\dot{y} = s - y, \quad \dot{s} = 1, \quad s(0) = 0,$$

respectively. The scalar equation $\dot{y} = t - y$ is linear nonhomogeneous, but its autonomous lift has the affine vector field

$$F(y, s) = (s - y, 1),$$

which is not linear because $F(0, 0) = (0, 1) \neq (0, 0)$.

Question: How does a repeated equilibrium affect a phase line?

Analyze

$$\dot{x} = (x + 1)^2(2 - x).$$

Solution. The equilibria are $x = -1$ and $x = 2$. Since $(x + 1)^2 > 0$ for $x \neq -1$,

x	$(-\infty, -1)$	-1	$(-1, 2)$	2	$(2, \infty)$
$f(x)$	+	0	+	0	−

and the phase line is

$$\longrightarrow \odot_{-1} \longrightarrow \bullet_2 \longleftarrow .$$

Thus -1 attracts from the left and repels from the right; it is half-stable and Lyapunov unstable. The point 2 is asymptotically stable. Moreover,

$$\lim_{t \rightarrow \infty} x(t) = \begin{cases} -1, & x(0) \leq -1, \\ 2, & x(0) > -1. \end{cases}$$

Since $f'(2) = -9$, perturbations near 2 decay to leading order as e^{-9t} . Since $f'(-1) = 0$, linearization does not classify -1 .

Question: How do phase line, potential, and basin agree?

For

$$\dot{x} = -x(x - 1)(x + 2),$$

find the equilibria, stability, basins, and a potential V satisfying $\dot{x} = -V'(x)$.

Solution. The sign table is

x	$(-\infty, -2)$	-2	$(-2, 0)$	0	$(0, 1)$	1	$(1, \infty)$
$f(x)$	+	0	−	0	+	0	−

Hence -2 and 1 are asymptotically stable, while 0 is unstable. Their basins are

$$\mathcal{B}(-2) = (-\infty, 0), \quad \mathcal{B}(1) = (0, \infty).$$

Expanding $-f(x)$ gives

$$-f(x) = x(x - 1)(x + 2) = x^3 + x^2 - 2x,$$

so one may take

$$V(x) = \frac{x^4}{4} + \frac{x^3}{3} - x^2.$$

Question: How do phase line, potential, and basin agree? (continued)

Along solutions,

$$\dot{V} = V'(x)\dot{x} = -f(x)^2 \leq 0.$$

The asymptotically stable equilibria are local minima of V ; the unstable equilibrium is a local maximum.

Visualization: Three local bifurcation signatures

With r as the parameter and x as the state,

type	normal form	equilibrium branches	structural condition
saddle-node	$\dot{x} = r - x^2$	$x = \pm\sqrt{r} \ (r \geq 0)$	no imposed symmetry or persistent branch
transcritical	$\dot{x} = rx - x^2$	$x = 0, \ r$	persistent branch $x = 0$
pitchfork	$\dot{x} = rx - x^3$	$x = 0, \ \pm\sqrt{r} \ (r \geq 0)$	$x \mapsto -x$ symmetry

Observation: Structural requirements for the three signatures

For unrestricted one-parameter scalar families, the generic equilibrium bifurcation is the saddle-node. Transcritical and pitchfork bifurcations are generic only within families that retain the indicated persistent branch or symmetry.

Question: Which local bifurcation occurs?

Find the equilibrium branches and their stability for

$$\dot{x} = \mu - (x - 1)^2, \quad \dot{x} = x(\mu - 2x), \quad \dot{x} = \mu x - 2x^3.$$

Solution. For $f_1(x; \mu) = \mu - (x - 1)^2$,

$$x_{\pm} = 1 \pm \sqrt{\mu} \quad (\mu \geq 0), \quad (f_1)_x = -2(x - 1).$$

For $\mu > 0$, x_- is unstable and x_+ is asymptotically stable. They collide at $(\mu, x) = (0, 1)$: a saddle-node. At $\mu = 0$, $\dot{x} = -(x - 1)^2$, so $x = 1$ attracts from the right and repels from the left: it is half-stable and Lyapunov unstable. For $\mu < 0$, there is no equilibrium.

For $f_2(x; \mu) = x(\mu - 2x)$,

$$x = 0, \quad x = \frac{\mu}{2}, \quad (f_2)_x = \mu - 4x.$$

The scalar linearized rates on the two branches are μ and $-\mu$, so the branches exchange stability at the origin: a transcritical bifurcation. At $\mu = 0$, $\dot{x} = -2x^2$, so the coincident equilibrium is half-stable and Lyapunov unstable, attracting from the right and repelling from the left.

For $f_3(x; \mu) = \mu x - 2x^3$,

$$x = 0, \quad x_{\pm} = \pm\sqrt{\frac{\mu}{2}} \quad (\mu > 0).$$

At the origin $(f_3)_x = \mu$, while

$$(f_3)_x(x_{\pm}) = \mu - 6\frac{\mu}{2} = -2\mu < 0.$$

Two asymptotically stable branches emerge as the origin loses stability: a supercritical pitchfork. At $\mu = 0$, $\dot{x} = -2x^3$ points toward the origin from both sides, so the origin remains asymptotically stable at the bifurcation value.

Question: What does an overdamped reduction forget?

For fixed initial data $y_0, v_0 \in \mathbb{R}$ and $\varepsilon > 0$, solve

$$\varepsilon y'' + y' = 1, \quad y(0) = y_0, \quad y'(0) = v_0.$$

Compare the solution with the reduced equation obtained by setting $\varepsilon = 0$.

Solution. Put $v = y'$. Solving $\varepsilon v' + v = 1$ with $v(0) = v_0$ and then integrating gives

$$\begin{aligned} v(t) &= 1 + (v_0 - 1)e^{-t/\varepsilon}, \\ y(t) &= y_0 + t + \varepsilon(v_0 - 1)(1 - e^{-t/\varepsilon}). \end{aligned}$$

The reduced solution is $y_{\text{red}}(t) = y_0 + t$. For every $t \geq 0$,

$$|y(t) - y_{\text{red}}(t)| \leq \varepsilon |v_0 - 1|.$$

Thus the positions converge uniformly on $[0, \infty)$ as $\varepsilon \downarrow 0$. For the velocities,

$$v(t) - y'_{\text{red}}(t) = (v_0 - 1)e^{-t/\varepsilon}.$$

At every fixed $t > 0$ this tends to zero, but at $t = 0$ it remains $v_0 - 1$. Unless $v_0 = 1$, the reduced equation therefore misses the initial velocity. The factor $e^{-t/\varepsilon}$ reduces the velocity discrepancy by e^{-1} every ε units of time, identifying the fast initial layer.

Question: How many decimal places does step refinement support?

An unknown quantity Q is approximated with step sizes $h, h/2, h/4$:

$$Q_h = 1.20384, \quad Q_{h/2} = 1.20496, \quad Q_{h/4} = 1.20524.$$

Assuming $p > 0$, $C \neq 0$, and

$$Q_h = Q + Ch^p + o(h^p) \quad (h \downarrow 0),$$

estimate p , the finest-grid error, and Q .

Solution. The ratio gives the estimated order:

$$\frac{Q_{h/2} - Q_h}{Q_{h/4} - Q_{h/2}} = \frac{0.00112}{0.00028} = 4 = 2^2 \Rightarrow p \approx 2.$$

The finest-grid error estimate and Richardson extrapolation are

$$\begin{aligned} |Q - Q_{h/4}| &\approx \frac{|Q_{h/4} - Q_{h/2}|}{2^2 - 1} = \frac{0.00028}{3} \approx 9.33 \times 10^{-5}, \\ Q &\approx Q_{h/4} + \frac{Q_{h/4} - Q_{h/2}}{3} = 1.205333 \dots \end{aligned}$$

The data support three decimal places, provided the asymptotic error model is valid and roundoff is negligible.